Capturing Reactive Textile Dyes Using Coconut-Shell Activated Carbon Alginate Beads: The Effects of Activated Carbon Concentration and Chitosan Coating

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Introduction

Textile dyeing is a major source of **industrial water pollution**, releasing large amounts of colored wastewater that can negatively impact aquatic ecosystems. **Reactive dyes**, such as Procion MX Turquoise (C.I. Reactive Blue 140), are especially challenging to remove because they are designed to remain stable and highly soluble in water. Although **activated carbon** is an effective adsorbent, recovering loose carbon after treatment can be difficult and costly.

Previous research has demonstrated that **sodium alginate beads** can immobilize activated carbon, allowing easier separation and reuse, while **chitosan coatings** may enhance removal of negatively charged dyes through electrostatic attraction. Building on these findings, we investigated how activated **carbon loading (0%, 10%, and 20%)** and **chitosan coating** influenced the removal of a reactive textile dye.

Our goal was to create a **biodegradable, low-cost, and recoverable adsorption system** that improves dye removal while maintaining the ability to separate the adsorbent from treated water.

Research Questions

Can biodegradable calcium alginate beads containing coconut-shell activated carbon provide an effective and recoverable method for removing reactive textile dyes from contaminated water, and how do carbon loading and chitosan coating impact their performance?

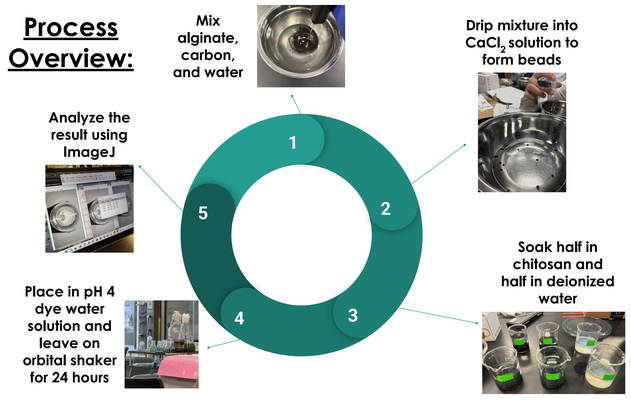
Methods and Materials

Materials:
- Sodium alginate
- Calcium chloride
- Coconut-shell activated carbon
- Chitosan
- White vinegar
- Sodium chloride
- Procion MX Turquoise dye (C.I. Reactive Blue 140)
- Deionized water

Equipment:
- Orbital shaker
- Beakers and flasks
- Steel bowls and containers
- Measuring spoons
- Scoopula
- Strainers
- .01 Digital scale
- Syringes
- Smartphone lightbox
- ImageJ software

A **2% sodium alginate gel** was mixed with **0%, 10%, or 20% activated carbon** and formed into beads using **calcium chloride crosslinking**. Half of each bead type was coated with **chitosan** in **pH 4 water**, while the remaining beads stayed uncoated. Six bead formulations were tested in **18 dye vials (3 replicates each)** containing **40 mL of dye solution** and shaken for **24 hours**. After treatment, **dye removal** was measured using standardized smartphone images analyzed with **ImageJ**. **Bead swelling** and **water absorption** were also measured to evaluate adsorption performance. A preliminary trial using **0.5 g of beads** was adjusted to **4 g per vial** after low dye removal was observed.

Process Overview:



Results & Discussion

Dye Removal Derived From ImageJ intensity

Formula: Dye darkness (treated) = -log10(I/255) Dye removed % = (control darkness - treated darkness) / control darkness x 100

Formulation	Coating	Carbon %	Photo brightness I (0-255)	Dye darkness (treated)	Dye darkness (control)	Dye removed (%)
0-U	Uncoated	0	188.068	0.1322	0.1586	**16.6**
10-U	Uncoated	10	208.227	0.0880	0.1586	**44.5**
20-U	Uncoated	20	240.545	0.0253	0.1586	**84.0**
0-C	Coated	0	186.177	0.1366	0.1586	**13.9**
10-C	Coated	10	190.05	0.1277	0.1586	**19.5**
20-C	Coated	20	205.316	0.0941	0.1586	**40.7**

We measured **dye removal** by photographing each vial and reading its **red-channel brightness (I)** in **ImageJ**. Brightness is converted to **absorbance** with **A = -log10(I/255)**, which by **Beer's Law** tracks dye concentration, and every vial is compared to the **dye-only control** to give **percent removed.** Removal rose with **carbon loading** and peaked at **84% for the uncoated 20% bead**.

Effect of Carbon Loading

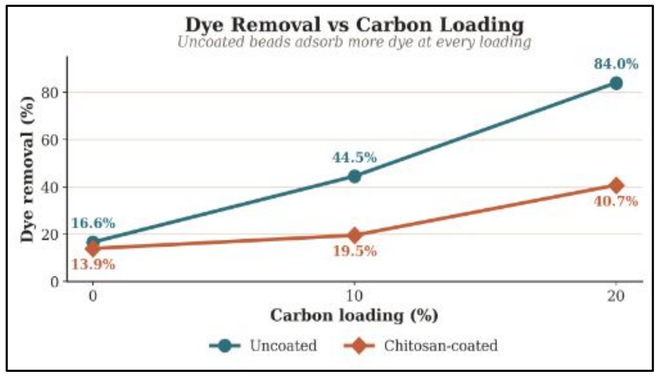
Increasing coconut-shell **activated carbon loading** significantly improved **dye removal**. Uncoated beads increased from **16.6% removal at 0% carbon** to **84.0% removal at 20% carbon**, demonstrating that **activated carbon** was the primary factor responsible for **dye adsorption**.

The carbon remained **trapped inside the alginate structure**, allowing the beads to be physically removed from treated water without releasing carbon particles. These results supported our hypothesis that **higher carbon loading** would improve **dye removal**.

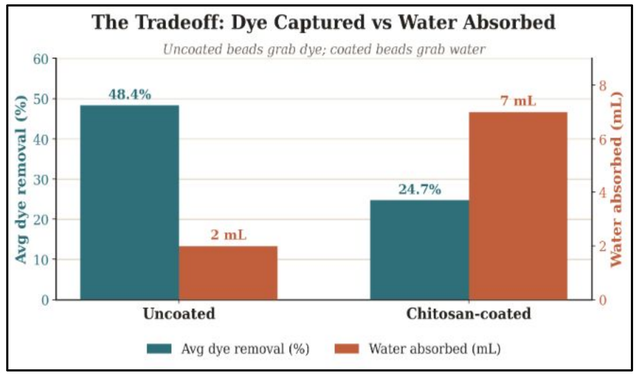
Effect of Chitosan Coating

Chitosan-coated beads performed worse than uncoated beads at every carbon loading. While chitosan was expected to improve adsorption through attraction between **positively charged chitosan** and **negatively charged dye molecules**, the coating created an unintended limitation.

Chitosan-coated beads absorbed significantly more water and experienced greater **swelling**, gaining approximately **65% mass** compared to **37% for uncoated beads**. They also absorbed approximately **7 mL of water** compared to **2 mL for uncoated beads**. This excessive swelling likely reduced **dye diffusion** into the bead and limited contact with the **activated carbon**.



Graph 1: Dye Removal vs Carbon Loading. Dye removal increased with activated carbon loading, while uncoated beads consistently outperformed chitosan-coated beads.



Graph 2: Dye Captured vs Water Absorbed. Uncoated beads achieved higher dye removal despite lower water absorption.

Dye Absorption vs. Water Uptake

The **selectivity index** helped distinguish **dye adsorption** from simple **water absorption**. Uncoated beads became increasingly selective as **carbon loading** increased, reaching a **selectivity index of 2.11 at 20% carbon**, while coated beads only reached **0.60**. These results indicate that **uncoated beads** effectively captured dye through **activated carbon adsorption**, while coated beads primarily absorbed water due to excessive **swelling**.

Selectivity index (dye captured per unit swelling)

Formula: Selectivity = Dye removal % / Swelling % (higher = the bead grabbed dye, not just water)

Formulation	Coating	Carbon %	Dye removal %	Swelling %	Selectivity index
0-U	Uncoated	0	16.6	32.5	**0.51**
10-U	Uncoated	10	44.5	37.0	**1.20**
20-U	Uncoated	20	84.0	39.8	**2.11**
0-C	Coated	0	13.9	62.3	**0.22**
10-C	Coated	10	19.5	66.5	**0.29**
20-C	Coated	20	40.7	68.3	**0.60**

The **selectivity index** divides each bead's **dye removal** by how much it **swelled**, so a high value means the bead pulled in **dye** rather than just soaking up **water**. **Uncoated beads** grew steadily more selective as **carbon increased** (up to **2.11**), while **chitosan-coated beads** stayed at or below **0.60**. This one number captures the study's main finding, **uncoated beads adsorb dye, coated beads mostly absorb water**.

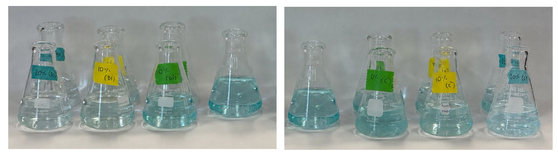
Conclusion

Coconut-shell activated carbon successfully improved **reactive dye removal** when immobilized inside **calcium alginate beads**. The **20% carbon uncoated beads** achieved the highest performance, removing **84% of the dye** while keeping the carbon contained within the bead structure for easier recovery.

Although **chitosan** was expected to increase dye capture through electrostatic attraction, the coating reduced performance by increasing **water absorption** and **bead swelling**. These results demonstrate that a simpler **uncoated high-carbon bead** provided the best balance between **adsorption efficiency**, stability, and recoverability.

This study shows that **biodegradable alginate-carbon beads** have potential as a **sustainable alternative** for treating **dye-contaminated wastewater** while reducing the challenges associated with recovering loose activated carbon.

After 24 Hours with Beads: Visible Reactive Dye Removal



Uncoated: 20%, 10%, 0%, Control **Chitosan coated:** Control, 0%, 10%, 20%

Future Directions

Future research will focus on optimizing **coating thickness**, testing **higher carbon loadings**, evaluating **bead reuse** over multiple adsorption cycles, and investigating how environmental factors such as **salinity and pH** affect adsorption. Additional studies could explore methods for safely destroying or regenerating captured dyes, including **photocatalytic** approaches using **titanium dioxide** to degrade collected dye molecules.

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